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# Secure Packages with CodeArtifact



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**Packages** [Info](#)  [Delete package](#) [View connection instructions](#)

< 1 2 3 > ⚙

|                       | Package name                             | Namespace                | Format | Latest version |
|-----------------------|------------------------------------------|--------------------------|--------|----------------|
| <input type="radio"/> | <a href="#">backport-util-concurrent</a> | backport-util-concurrent | maven  | 3.1            |
| <input type="radio"/> | <a href="#">classworlds</a>              | classworlds              | maven  | 1.1            |
| <input type="radio"/> | <a href="#">google</a>                   | com.google               | maven  | 1              |
| <input type="radio"/> | <a href="#">jsr305</a>                   | com.google.code.findbugs | maven  | 2.0.1          |
| <input type="radio"/> | <a href="#">google-collections</a>       | com.google.collections   | maven  | 1.0            |



# Introducing Today's Project!

In this project, I will demonstrate DevOps skills. I'm doing this project to learn about devops tools like Code Artifact

## Key tools and concepts

Services I used were EC2, CodeArtifact, IAM roles, IAM Policy, and such. Key concepts I learnt include using EC2, attaching roles and policies to it, CodeArtifact, Policies, security, etc.

## Project reflection

This project took me approximately an hour. The most challenging part was removing errors. It was most rewarding to share it with my connections.

This project is part three of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project within 3 days.





# CodeArtifact Security

## Issue

To access CodeArtifact, we need security. I ran into an error when retrieving a token because I didn't provide a permission to it.

## Resolution

To resolve the error with my security token, I created IAM policies and role for EC2. This resolved the error because it gives permission to access.

It's security best practice to use IAM roles because it requires certain permissions and policies and work accordingly.





## The JSON policy attached to my role

The JSON policy I set up grants authorization, endpoints, and read from repository in CodeArtifact.

```
1 {  
2   "Version": "2012-10-17",  
3   "Statement": [  
4     {  
5       "Effect": "Allow",  
6       "Action": [  
7         "codeartifact:GetAuthorizationToken",  
8         "codeartifact:GetRepositoryEndpoint",  
9         "codeartifact:ReadFromRepository"  
10      ],  
11      "Resource": "*"   
12    },  
13    {  
14      "Effect": "Allow",  
15      "Action": "sts:GetServiceBearerToken",  
16      "Resource": "*",  
17      "Condition": {  
18        "StringEquals": {  
19          "sts:AWSServiceName": "codeartifact.amazonaws.com"  
20        }  
21      }  
22    }  
23  ]  
24 }
```



# Maven and CodeArtifact

To test the connection between Maven and CodeArtifact, I compiled my web app using settings.xml

The settings.xml file configures Maven to set up this configuration and work with CodeArtifact.

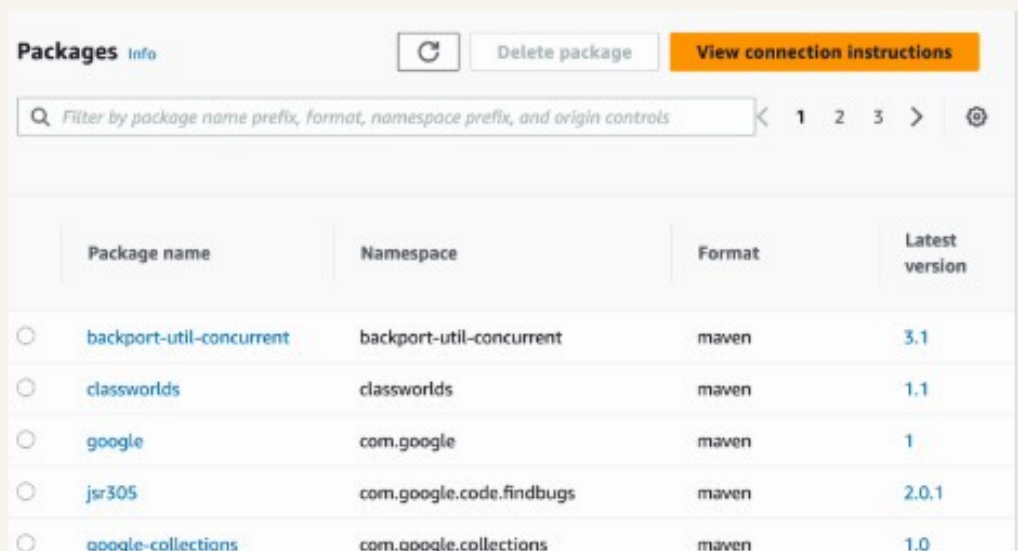
Compiling means fetching and assembling all files .

```
src > main > webapp > settings.xml
1  <settings>
2  <servers>
3  <server>
4  <id>nextwork-nextwork-devops-cicd</id>
5  <username>aws</username>
6  <password>${env.CODEARTIFACT_AUTH_TOKEN}</password>
7  </server>
8  </servers>
9  <profiles>
10 <profile>
11 <id>nextwork-nextwork-devops-cicd</id>
12 <activation>
13 <activeByDefault>true</activeByDefault>
14 </activation>
15 <repositories>
16 <repository>
17 <id>nextwork-nextwork-devops-cicd</id>
18 <url>https://nextwork-120569621470.d.codeartifact.ap-southeast-1.amazonaws.com/maven/nextwork-devops-cicd</url>
19 </repository>
20 </repositories>
21 </profile>
22 </profiles>
23 <mirrors>
24 <mirror>
25 <id>nextwork-nextwork-devops-cicd</id>
26 <name>nextwork-nextwork-devops-cicd</name>
27 <url>https://nextwork-120569621470.d.codeartifact.ap-southeast-1.amazonaws.com/maven/nextwork-devops-cicd</url>
28 <mirrorOf>*</mirrorOf>
29 </mirror>
30 </mirrors>
31 </settings>
32
```



# Verify Connection

After compiling, I checked settings.xml. I noticed there are now many repositories inside it.



The screenshot shows a web interface titled "Packages" with a sub-header "Info". It includes a search bar with the placeholder text "Filter by package name prefix, format, namespace prefix, and origin controls". To the right of the search bar are buttons for "Delete package" and "View connection instructions". Below the search bar is a table with four columns: "Package name", "Namespace", "Format", and "Latest version". The table lists five packages: "backport-util-concurrent", "classworlds", "google", "jsr305", and "google-collections". Each package has a radio button to its left, indicating it is selected.

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